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Project Summary

Medical professionals have always faced the challenge of writing comprehensive medical reports for each patient. They now have to deal with more administrative work as a result of this time-consuming process, possibly leading to burn out. Additionally, manual report generation is more prone to errors including typographical mistakes and misinterpretations, and so these errors could negatively affect patient care and safety. Therefore, we aim to develop a system that helps in improving the generation of accurate and well-structured reports based on the processing of radiology scans. We present a region guided radiology report generation system that uses object detection to detect individual anatomical regions in a chest x-ray image and generates a report describing the most salient regions using cutting-edge deep learning technologies. After analyzing and detecting the regions, we use a Vision Language Model (VLM) pretrained specifically on medical data that takes into consideration the regions of interest and focuses on them while generating a report that is more medically accurate and relevant. The automation of the manual generation process can result in faster diagnosis and increased efficiency. So, by employing this approach we can assist radiologists and optimize the process of automatic report generation for medical images. Furthermore, the system can lessen the administrative workload on the medical professionals, decreasing the medical healthcare costs. Overall, our system will address a wide range of critical issues in chest radiology and patient care.

Keywords: Report generation, Region-guided, Deep learning, Chest X-ray, Vision Language Model

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List of **Abbreviations**

- NLP: Natural Language Processing
- NLG: Natural Language Generation
- CNN: Convolutional Neural Networks
- LSTM: Long Short-Term Memory
- OOP: Object Oriented Programming
- RPN: Region Proposal Network
- ROI: Region Of Interest
- IOU: Intersection Over Union
- VLM: Vision-Language Model
- LLaVA: Large Language-and-Vision Assistant
- LLaVA-Med: Large Language-and-Vision Assistant for Medical reports

Chapter 1

Introduction

1.1. Background:

Radiology plays a crucial role in modern healthcare, offering a range of examinations like x-rays, ultrasound, CT scans, and MRI scans that provide valuable insights into a patient's anatomy without the need for invasive surgeries. These imaging techniques aid in diagnosing medical conditions and monitoring treatment effectiveness [1]. Traditionally, radiologists manually create reports based on their interpretations, which is a time-consuming process prone to errors. To reduce this burden, researchers and technologists have combined deep learning and NLP to automate the generation of medical reports from radiology images. This innovative approach involves analyzing and translating the images to extract meaningful information that can guide diagnosis and treatment decisions.

In particular, vision-language models (VLMs) have emerged as a powerful tool in this domain. VLMs integrate visual data with textual data, allowing for the simultaneous processing and understanding of images and their corresponding descriptions. By leveraging VLMs, the automation of report generation for x-rays and other imaging modalities becomes more sophisticated, producing high-quality, coherent reports that mirror the insights of experienced radiologists. Furthermore, the application of vision-language models and NLP techniques in radiology not only enhances the efficiency and accuracy of report generation but also facilitates better clinical decision-making. Automating the report generation process for x-rays allows radiologists to dedicate more time to analyzing complex cases and providing personalized patient care [1]. Embracing these advancements in radiology automation enhances efficiency and accuracy in healthcare practices.

1.2. **Motivation:**

With the ever-growing demands on healthcare practitioners to deliver individualized medical reports for every patient, they are under increasing pressure which could lead to burnout. This increasing pressure is a significant concern, as the physical and mental well-being of healthcare professionals is crucial for maintaining high standards of patient care. Moreover, the increased labor required for manual report generation can lead to additional costs being passed on to patients or healthcare systems, potentially resulting in higher healthcare expenses.

In addition to that, manual report generation is more prone to errors including typographical mistakes and misinterpretations. These errors can have serious implications, potentially leading to incorrect diagnoses or inappropriate treatment plans, which could negatively affect patient care and safety. Ensuring accuracy in radiology reports is essential for effective clinical decision-making and optimal patient outcomes.

Therefore, this research aims to contribute to the field of radiology by generating automatic medical reports with high accuracy. By leveraging advanced technologies such as vision-language models (VLMs) and natural language processing (NLP), our approach seeks to enhance the reliability and efficiency of report generation. These automated systems can significantly reduce the workload on radiologists, allowing them to focus more on complex cases and direct patient interactions, ultimately improving the quality of care provided.

We believe our system would be beneficial to hospitals, radiology centers and companies that manufacture radiology machines, as if our system was to be installed on the machines, it could speed up the process of generating reports and handing them out to the patients. This increased efficiency would not only enhance patient satisfaction by reducing wait times but also improve the overall workflow within healthcare facilities.

1.3. Objectives:

We aim to develop a system that improves the generation of accurate and well-structured reports based on the processing of chest X-ray scans. By using Vision Language Models (VLMs) and Natural Language Processing (NLP), our system will automatically analyze and interpret X-ray images, producing detailed medical reports. This system will be able to positively impact the healthcare industry. The automation of the manual generation process will result in faster diagnosis and increased efficiency. Moreover, it offers the opportunity for earlier disease detection, allowing treatment at a stage when it is more likely to be effective. Employing this system makes the medical reports less prone to human errors, reducing the risk of misinterpretation. Furthermore, the system will lessen the administrative workload on the medical professionals, decreasing the medical healthcare costs, making services more affordable and accessible. Overall, our system will address a wide range of critical issues in chest radiology and patient care. In addition to that, we aim to develop an easy-to-use interface so that it doesn't require a lot of time from the radiologist to get familiar with our system.

1.4. Scope:

The project involves the development and implementation of a Region-Guided Radiology Report Generation System for chest x-ray images. The system is designed to automate the analysis of specific anatomical regions and generate detailed radiology reports. However, the boundaries and limitations will be stated along with what is included in our system.

1.4.1 Boundaries and Limitations:

1. The system focuses exclusively on chest x-ray images and does not extend to other types of medical imaging (e.g., MRI, CT scans).

2. The analysis is limited to predefined anatomical regions within the chest.
3. The generated reports are intended for use by radiologists and medical practitioners, not directly by patients
4. Manual review and verification by radiologists are necessary to ensure the clinical validity and accuracy of the reports.
5. The system's capabilities are confined to the detection and description of abnormalities within the chest x-ray images; it does not provide diagnostic conclusions or treatment recommendations.

1.4.2 System Inclusions:

1. Preprocessing of chest x-ray images.
2. Detection of anatomical regions within chest x-rays.
3. Report Generation.
4. User Interface.

1.5. Significance of the Study:

To the best of our knowledge this is the first study to use a region guided approach along with a medically pretrained vision language model to generate medical reports for chest x-rays. Most of the conventional methods rely on image level features to generate the reports, while the method we propose places a big emphasis on individual regions which has the ability to improve the accuracy and the relevance of radiographic findings. As the model detects different regions inside the chest, analyzes them, then targets the most important regions and passes them to our vision language model to generate a report that is more focused on the important regions.

The use of vision-language models, particularly those pretrained on medical data like the LLaVA-Med model[2], greatly enhances the system's effectiveness. Medically pretrained models have a good understanding of medical terminology and diagnostic contexts. These models are also specifically designed to handle the complexities of medical imaging and report generation, as it integrates visual and textual information seamlessly, enabling it to produce highly accurate and clinically relevant reports.

This can be of great assistance to healthcare professionals, as it can serve as a useful clinical decision support tool and can assist them in locating important findings quickly which can result in quicker intervention and better patient outcomes.

1.6. Outline the structure of the report:

The report will be structured as follows; chapter 1 contains an introduction to the project including the problem statement and the objective. Chapter 2 contains details about previous work and literature review. Chapter 3 explores the description, requirements, and design of the system. Chapter 4 digs into the implementation, structure of the code and the user interface. Chapter 5 contains our quantitative and qualitative results and the discussion of these results. Chapter 6 includes the conclusion and future work.

Chapter 2

Related Work

2.1 Introduction to Literature Review:

The purpose of this literature review is to get a better understanding of the existing research and debates relevant to the automated generation of radiology reports. Important concepts, research methods and different experimental models will be learnt to highlight research gaps that future work should focus on.

In this chapter, we will discuss how the AI driven report generation has developed, what methods were conducted, the key findings and what can be improved or added.

2.2 Historical Perspective:

Automated radiology report generation has evolved from early attempts to a sophisticated system today. The ability to extract information from medical images accurately and efficiently has been made possible by developments in machine learning and natural language processing (NLP). While deep learning methods have proven effective for picture classification and captioning, radiology report creation still faces difficulties in understanding and connecting complex medical visual contents with precise natural language descriptions.

AI driven report generation is an important matter to continue conducting research on as it can improve efficiency and accuracy. This can lead to faster diagnoses, reduced healthcare costs, and improved patient outcomes, particularly in settings with limited access to radiologists.

2.3 Theoretical Framework:

Recent research on automated radiology reports generation from chest X-rays has predominantly revolved around using deep learning models, particularly convolutional neural networks (CNNs), which have demonstrated high efficiency in detecting a range of abnormalities. The availability of public datasets has encouraged more research in the field of chest x ray report generation.

Furthermore, the utilization of transfer learning, wherein pre-trained models are adapted to chest X-ray datasets, has emerged as a beneficial strategy to address constraints related to limited annotated data and computational resources. Additionally, efforts have been directed towards developing multi-label classification models capable of concurrently identifying and localizing various pathologies within chest X-ray images.

Current research focuses on identifying common abnormalities found in chest X-rays, like pneumonia and lung nodules. Nevertheless, there's a lack of reliable algorithms designed to detect uncommon abnormalities and challenging cases. Another research gap is that most researchers use image level features without paying attention to the specific regions of interest.

Traditionally, RNNs were used as sequence-to-sequence models[3]. However, attention mechanisms used in transformers as introduced by Vaswani, A. et al[4] have revolutionized the state of natural language processing. Transformers outperform sequential models, like RNNs, because they better understand the context of text, and therefore can generate more relevant results. Additionally, Large language Models (LLMs) trained on huge textual data enable LLMs to have “knowledge”, similar to that of a human, which it can effectively apply to various settings, including report generation.

2.4 Previous Research and Studies:

Different approaches have been introduced for classifying chest x-rays and generating automated reports like the integration of CNN and LSTM for detecting abnormalities followed by attention based mechanism for report generation[5], double feature transformers which uses two CNN backbones to extract image features and then pass them to the encoder[6], Posterior-and-Prior Knowledge Exploring-and-Distilling approach (PPKED) which tries to imitate the working pattern of the radiologists who will first identify the abnormal regions, give them tags or captions then use their experience to write the report[7], a multi-view image fusion which incorporate the features extracted from the frontal and lateral view of the same patient and pass them to the encoder[8] and AMLMA approach which uses multi- level feature extraction and multiple attention mechanisms for report generation task[9].

2.5 Current State of the Field:

The field of automatic radiology report generation from chest X-rays has made notable strides with deep learning techniques, achieving high accuracy in detecting common abnormalities like pneumonia. Recent trends include attention mechanisms and transfer learning, using vision language models and utilizing conversational assistants specialized in the medical field, and enhancing model performance and generalization.

In recent years, Vision Language Models (VLMs), which integrate visual and textual data processing, have also shown significant promise in various domains, including computer-aided medical diagnosis. These models can synthesize the detailed information contained in medical images and accompany radiology reports to enhance diagnostic accuracy. Suhyeon Lee et al. [10] present in a study a novel approach that fine-tunes pre-trained LLMs to read and generate CXR images using a Vector Quantized Generative Adversarial Network (VQ-GAN) framework. This method embeds image tokens into the same latent space as text tokens, allowing the model to treat images as sequences of text tokens. This integration facilitates seamless multimodal tasks, enabling the model to generate text from images (CXR-to-Report) and images from text (Report-to-CXR) without altering the LLM's structure or training objectives.

Vision Language Models offer high flexibility in converting text to images and images to text. They build upon large language models in that they can comprehend images. Another example is MedXChat [11] a framework designed to support various chest X-ray (CXR) tasks such as report generation, Visual Question Answering (VQA), and text-to-CXR generation. It can produce reports for medical images, due to its ability to deal with images and to extract relevant features. Furthermore, it can generate medical images given particular instructions as a prompt. MedXChat uses advanced visual encoders and large language models (LLMs) to translate medical images into language-like features. It incorporates the Stable Diffusion (SD) architecture to create detailed medical images from text prompts by following specific instructions. This framework allows for smooth interaction between different tasks in a single environment, enhancing its clinical effectiveness.

Additionally, LLaVA-Med is one of the latest advancements in the area of large language vision models. A multimodal conversational assistant specifically designed for biomedical applications. This innovative model leverages a large-scale biomedical figure-caption dataset and a curriculum learning approach to address the complex demands of biomedical image understanding and conversation. The training pipeline, which utilizes GPT-4 to generate instructions from text, enables the model to produce comprehensive instruction-following data without manual annotations.

Finally, one of the key studies that referenced the decision for our pipeline is the paper “Interactive and Explainable Region-guided Radiology Report Generation” [12]. The paper presents a model that aims to improve the automatic generation of radiology reports by focusing on anatomical regions within medical images. The proposed method detects specific anatomical regions and generates descriptive sentences for each, ensuring that the generated report is comprehensive and clinically accurate. Unlike previous approaches that relied on image-level features, this region-guided model enhances transparency and allows for interactive capabilities, enabling radiologists to select regions of interest and manually draw bounding boxes for targeted descriptions. The study demonstrates the model's effectiveness through extensive experiments, outperforming state-of-the-art models in both report generation and clinical relevance. This approach not only improves the accuracy and completeness of the reports but also provides a higher degree of explainability, fostering trust and usability in clinical settings.

Chapter 3

Materials and Methods

3.1 System Description:

3.1.1 Context of the System:

The system is a region-guided Radiology Report Generation system designed to analyze chest x-ray images and generate reports focusing on individual anatomical regions. Its context includes the entire process from receiving the x-ray images as an input to producing the reports. Additionally, the system may interact with external components like hospitals' systems to store the reports in the patients' files.

3.1.2 Objectives and Requirements:

Objective: Develop a system that accurately detects anatomical regions in radiological images and generates detailed reports focusing on these regions.

Requirements:

1. Implement machine learning algorithms for accurate detection and segmentation of anatomical regions.
2. Implement 2 binary classifiers to select the regions of interest that would be featured in the report.
3. Use the VLM model to analyze the chest x-ray image while focusing on the selected regions to generate a detailed radiology report.

3.2 System Requirements:

The general functionality of the system is taking the image of the chest x-ray as an input, analyzing it, detecting the anatomical regions inside the chest, then producing the report.

The functional requirements for the system include:

1. **Title:** Upload X-ray Image

Description: The user will be able to choose and upload from his device the chest x-ray image that he wants to generate the report for by using the “Upload” button.

2. **Title:** Generate the Report

Description: The system will generate the report for the uploaded image when the user clicks on the button “Generate”.

The non-functional requirements for the system include:

1. **Title:** Security

Description: Maintain data integrity and privacy during the training and prediction stages.

2. **Title:** Reliability

Description: Generate accurate and reliable results by the system.

3. **Title:** Maintainability

Description: Implement organized clear structure and documentation to facilitate changes and maintenance.

3.3 Design Constraints:

Constraints for the system may include:

1. **Usability:** Designing a user-friendly interface that is easy to use while considering the needs of medical professionals.
2. **Expandability:** Ensuring that the system is scalable and can handle future changes.

3.4 Research Design:

Our plan for conducting the project began with an in-depth exploration of previous work done in the same area. This initial stage involved performing an extensive survey of academic papers and studies that discussed similar ideas, focusing on summarizing key findings from past research. During this literature review, we aimed to identify the strengths and limitations of existing approaches and identify any research gaps that could be addressed by our project. This foundational work provided us with a solid understanding of the current state of the field and informed the direction of our subsequent efforts.

In addition to this literature review, we began researching various VLMs to choose a model that is well suited with our objective to generate more accurate results.

Following the selection of the appropriate model, our next step was to collect a comprehensive dataset. We focused on gathering datasets that contain a wide range of radiological images and their corresponding annotations. These datasets are crucial for training and testing the model, ensuring it can accurately analyze and interpret medical images.

Once we found the suitable datasets, we moved on to the planning phase of the system implementation. This stage involved designing the overall architecture of the system and selecting the most suitable technologies and frameworks for implementation. A system architecture was developed, outlining the workflow from data preprocessing to model training and evaluation.

The primary steps taken to construct our pipeline are:

1. Preprocessing: Apply the necessary preprocessing functions for the chest x-ray images received like resizing to 265x265.
2. Train Object Detector: Construct an object detector model that detects the separate regions inside the chest, and trains it using labeled data.
3. Train Binary Classifiers: Train 2 binary classifiers that predict if a certain region is abnormal and if it is selected for sentence generation.
4. Load the VLM: Initialize necessary libraries and configurations, load model and processor with the specified configurations.
5. Apply Prompt Engineering: Use few-shot prompting technique to design a prompt to produce more accurate reports by crafting specific questions and instructions that guide the model to focus on important details.
6. Use the VLM: Pass the designed prompt to the model and use it to generate the chest x-ray report.
7. Evaluate model: Use Natural Language Generation (NLG) metrics in addition to Clinical Efficacy (CE) metrics to evaluate the model.

3.5 Architectural Design:

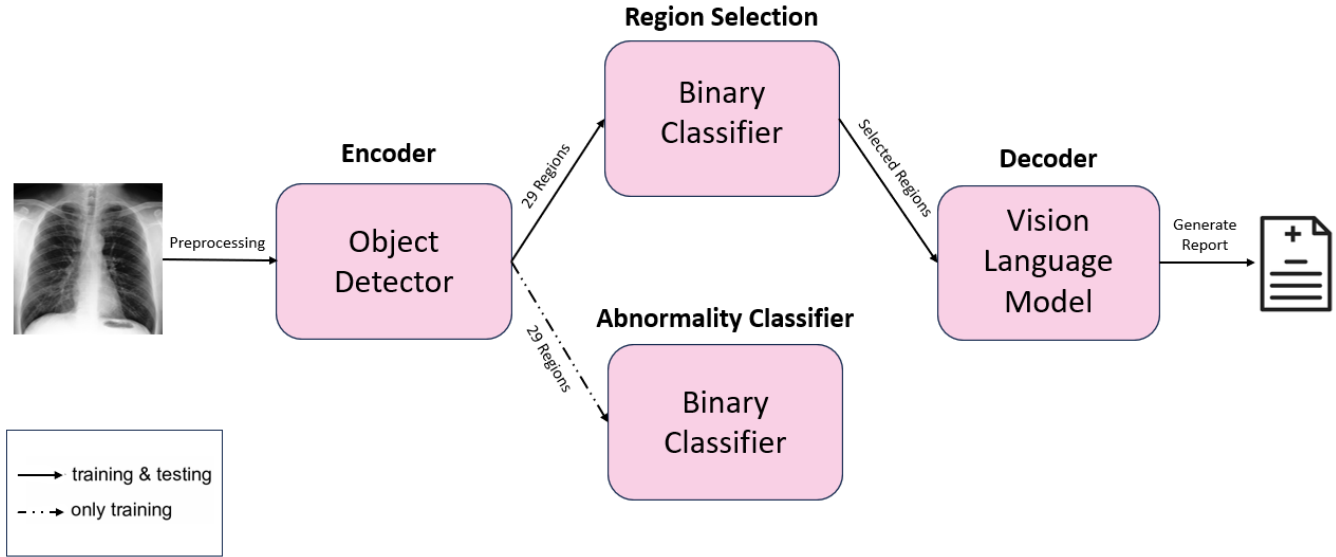


Figure 1: System Architecture

The Medical image is preprocessed and then passed to the object detector which gathers visual features for 29 individual chest anatomical regions. Two binary classifiers encode important information in the features regarding its abnormality, and choose key region features to pass them to the VLM to generate the report.

The system architecture begins with the acquisition of a chest x-ray image, which serves as the initial input for the entire process. The first stage involves applying a series of preprocessing functions to the raw image. These preprocessing steps are crucial for enhancing the quality of the x-ray and preparing it for the next steps. Preprocessing functions that are used include resizing, data augmentation and normalization, among other functions as well.

Once the preprocessing is complete, the refined chest x-ray image is then fed into an object detection model. This object detector is specifically designed to recognize 29 distinct anatomical regions within the chest area. These regions encompass various critical parts such as the lungs, spine, and diaphragm, ensuring comprehensive coverage of the chest anatomy.

After the object detector has successfully identified and segmented the anatomical regions, the next phase involves two specialized binary classifiers. The first classifier is to detect abnormal regions within the chest x-ray. It examines each of the 29 regions to identify any anomalies or signs of disease. The output from this classifier is a binary indication of whether each region is normal or abnormal.

The second binary classifier focuses on selecting the regions that should be included in the final radiology report. This classifier evaluates the importance and relevance of each anatomical region in the context of the overall diagnosis and ensures that only the most important regions are highlighted in the report.

Following the selection of the relevant regions, these regions are then passed on to the VLM. This model leverages natural language processing and medical imaging to generate a comprehensive and coherent radiology report. The VLM synthesizes information from the selected regions, along with the medical information the model is previously pretrained on, to produce a report that accurately reflects the findings from the chest x-ray.

3.6 Component Design:

Our architecture contains three main components which are the object detector, region selection and abnormality classifiers and the vision language model.

1. Object Detector

For object detection, we employ faster R-CNN model with a ResNet-50 backbone that has been pre-trained on ImageNet dataset. The Faster R-CNN framework is composed of a region proposal network (RPN) which is responsible for generating object proposals, i.e., bounding boxes that likely contain anatomical regions of interest. These proposals are created based on the feature maps that the ResNet-50 backbone extracts from the input chest x-ray image.

The other component of Faster R-CNN is the region of interest (RoI) pooling layer. This layer maps each object proposal onto the backbone’s feature maps and extracts smaller feature maps of a consistent spatial size for each proposal. This uniformity is essential for subsequent processing steps. Each of these RoI feature maps is then classified into one of the 29 predefined anatomical region classes, as well as an additional background class, following the standard procedures established in Faster R-CNN.

To extract the visual features for each of the 29 anatomical regions, we first identify the “top” object proposal for each region class. The top object proposal for a given region class is defined as the proposal with the highest probability score for that class out of all possible classes, and also the highest score among all proposals where the same region class is the top-1 class. If a region class does not achieve the highest score for any proposal, it is labeled as an undetected class. These undetected classes are automatically excluded by the region selection module, ensuring that only the most relevant regions are considered.

Subsequently, we obtain the region visual features, which are represented as a matrix with dimensions $R^{29 \times 1024}$. This is achieved by taking the 29 RoI pooling layer feature maps corresponding to the 29 top object proposals, each having dimensions $R^{29 \times 2048 \times H \times W}$. We then apply 2D average pooling over the spatial dimensions to condense the information, followed by a linear transformation that reduces the dimensionality from 2048 to 1024.

2. Region Selection and Abnormality Classifiers

The binary classifiers within these modules are composed of Multilayer Perceptrons (MLPs) designed to progressively reduce the input features down to a single output logit. These classifiers serve two main purposes: one classifier predicts whether a region should be selected for sentence generation, and the other determines if a region is abnormal. This approach ensures precise and efficient identification of the most relevant and critical regions for detailed analysis and reporting.

3. Vision Language Model (VLM)

The VLM is a crucial component of our system. Specifically, VLMs tailored for medical applications and pre-trained on extensive medical datasets, enabling them to accurately interpret and analyze medical images by combining visual and textual information to generate detailed and precise radiology reports. So, the integration of the VLM significantly enhances the overall capability of our system in producing high-quality medical reports.

3.7 Data Design:

The primary dataset that we used is “Chest Imagenome” [13] dataset that was derived from “MIMIC-CXR” dataset [14][15]. “MIMIC-CXR” dataset is a publicly available repository of chest radiographs and associated reports. This extensive dataset comprises 377,110 images and 227,827 reports. However, the “Chest Imagenome” dataset offers a unique perspective by incorporating combinations of relation annotations. These annotations are constructed as scene graphs for each image, capturing relationships between 29 chest X-ray anatomical locations and their corresponding attributes. “Chest Imagenome” dataset includes 242,072 images and their annotations. These annotations are constructed as scene graphs for each image, capturing relationships between 29 chest X-ray anatomical locations and their corresponding attributes. “Chest Imagenome” dataset includes 242,072 images and their annotations. This dataset enhances the depth of analysis by providing detailed anatomical insights and attributes associated with the radiographic images. Due to limited computational power, we were only able to use a sample of around 40,000 images of the original dataset to train, validate and test our model.

3.8 Algorithmic Design:

Multiple deep learning approaches are used in the project to achieve its objectives, including faster R-CNN and ReNet-50, both utilized within the object detector. Additionally, we employ a divide and conquer strategy to enhance the precision of the VLM. Instead of feeding the entire chest x-ray image into the model, we segment the image into 29 distinct anatomical regions. By dividing the image into smaller, more manageable regions, we enable the model to concentrate on the critical areas that require detailed analysis and reporting. This targeted approach ensures that the VLM can generate more accurate and precise reports.

3.9 Data Flow Diagram:

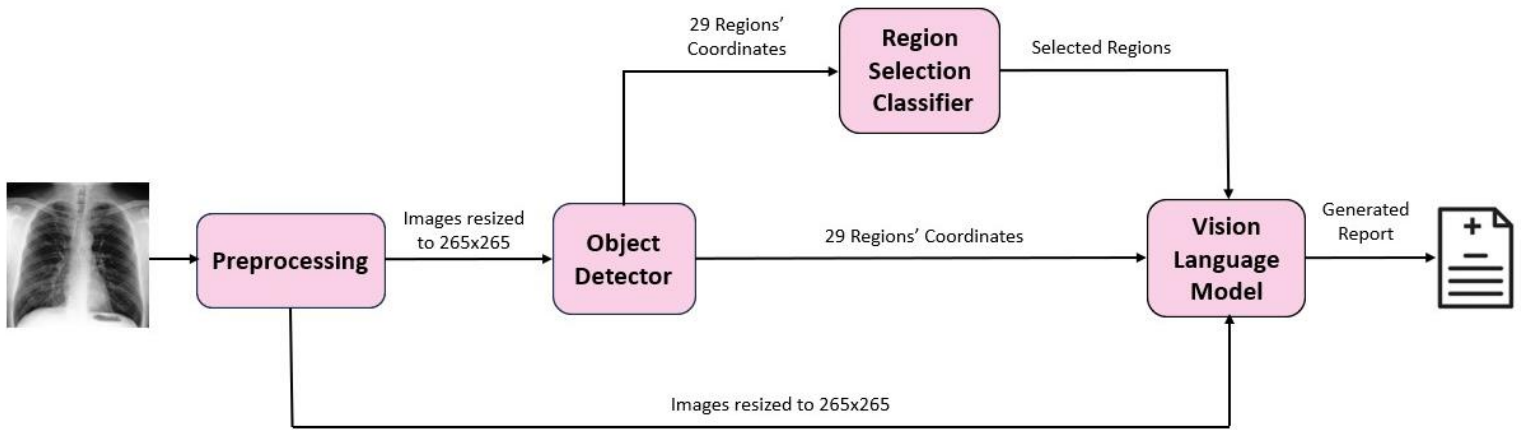


Figure 2: Data Flow Diagram

The system is given the chest x-ray image as input, then the image is preprocessed and resized to 265x264. The object detector then receives the preprocessed image as an input and detects the 29 regions inside the chest and generates an output that contains the coordinates of each of the 29 regions. These coordinates are passed to the binary classifier which selects the regions that should be focused on in the report. Then the VLM receives an input that consists of; the preprocessed image, the coordinates of each region, and the regions selected for the report.

3.10 Experimental Setup:

The configuration details for training our model on a sample of images include:

Table 1: Configuration details for training

Configuration	Value
Seed	41
image_input_size	265
percentage_of_train_set_to_use	0.5
percentage_of_val_set_to_use	1.0
batch_size	4
effective_batch_size	117
num_workers	8
epochs	20
lr	0.001
evaluate_every_k_steps	1000
patience_lr_scheduler	5
threshold_lr_scheduler	0.001
factor_lr_scheduler	0.5
cooldown_lr_scheduler	5

Chapter 4

Implementation and UI

4.1 Programming Languages, Tools and Techniques:

1. Python: Because of its flexibility and extensive libraries, it's widely used in the field of machine learning and AI.
2. Object Oriented Programming (OOP): For code organization and abstraction of complex systems into simpler, easier-to-understand elements.
3. Neural networks: An essential component of deep learning algorithms that is able to successfully recognize complicated links and patterns in data.
4. Pytorch: A commonly used deep learning framework that can efficiently train neural networks on large datasets, it also supports many deep learning tasks like object detection or image classification.
5. OpenCV: A powerful library for image processing and computer vision tasks.
6. Few-shot prompting: A prompt engineering technique to improve the performance of AI models by feeding them with a few examples to guide their responses and enhance their understanding of the task.
7. Fast API: A framework used for their high performance, ease of use, and ability to quickly build and deploy RESTful APIs.
8. Torch: A scientific computing framework that provides wide support for machine learning algorithms and is commonly used as the core library in Pytorch for implementing and training neural networks.

4.2 Code Structure:

This section outlines the structure of the code developed for this project. It provides an organized overview of the various components and modules, explaining their functionalities and interactions. The project's code is implemented using OOP in python and divided into folders. Each model is implemented in a class or multiple classes that contain multiple functions.

To start off, we first loaded all the requirements and dependencies that we will need. Then, our project starts with preparing the dataset format, implementing the object detector module, training it and saving its best weights, then implementing the 2 binary classifiers and using the object detector best weights to train them. After that, we implement our code for loading and using the VLM. In our case we use the LLaVA-Med model as our VLM. Lastly, we implement our pipeline to be ready for the deployment.

Below is a detailed explanation of each class and its functionality

1. “create_dataset”

We use the “ImaGenome” dataset which saves the data in Json files format, so we had to change it so that we can use it in the further steps. After creating the csv of the dataset, the attributes of the data were:

subject_id: ID of the patient whose image is used

study_id: ID of the study of that patient (since a patient can have several studies done to document the progression of a disease etc.)

image_id: ID of the single image

mimic_image_file_path: File path to the jpg of the single image

bbox_coordinates: A nested list that store the 4 coordinates of each of the existing chest anatomical regions

bbox_labels: A list that has the existing regions' labels corresponding to the bbox_coordinates.

The dataset is then split into train, validation and test datasets. For the validation and testing datasets, we add the `reference_report` attribute.

2. “preprocessing”

The image preprocessing step involves several important operations to prepare the images for model input. Initially, the images are resized to a 265x265 resolution while maintaining their original aspect ratio. This resizing ensures that the images fit into a consistent size without distorting the content, which is crucial for maintaining the integrity of the features within the image. After resizing, the images undergo normalization, which adjusts the pixel values to have a zero mean and unit standard deviation. This normalization process helps in stabilizing and accelerating the training process by ensuring that the data distribution is consistent across the dataset. Additionally, data augmentation techniques are applied to the images to enhance the robustness and generalizability of the model. These techniques include applying color jitter, Gaussian noise, and affine transformations.

3. Object Detector Module:

For the object detector, we use Faster R-CNN with a ResNet-50 backbone pre-trained on ImageNet. Faster R-CNN consists of a region proposal network (RPN) and region of interest (ROI) pooling layer.

3.1. “custom_rpn”

The class “CustomRegionProposalNetwork” inherits from PyTorch's “RegionProposalNetwork” but overrides its behavior to compute losses in both training and evaluation modes. It takes the image, extracted features and the target (if provided) as inputs and it generates

anchor boxes, obtains objectness scores and bounding box deltas, decodes deltas into proposals, filters proposals based on scores and image sizes and if targets are provided – targets are provided only in the training of the model –, computes objectness and bounding box regression losses. This class returns the proposed boxes and if it is in training mode, it also returns the losses.

3.2. “custom_roi_heads”

This class extends the RoIHeads class from PyTorch's torchvision library. It validates the target data types if provided, selects training samples if in training mode, applies RoI pooling to generates feature maps for the proposals, uses a box head network to extract feature vectors, predicts class scores and box regressions with a box predictor network. During training, it computes classification and box regression losses. Depending on the mode and configurations, it returns top region features, detections, class detections.

3.3. “object_detector”

This class inherits from nn.Module and implements Faster R-CNN model with a ResNet-50 backbone and a custom ROI heads and RPN (Region Proposal Network). The ResNet-50 backbone is modified to accept grayscale input images, so the first convolutional layer is changed to accept 1 input channel instead of 3. This class is used to map each object proposal onto the backbone feature maps, extract smaller feature maps of consistent spatial dimensions for the proposals. One of the 29 anatomical region classes is assigned to these ROI feature maps.

4. “binary_classifier_region_abnormal”

This class is designed to classify whether specific regions in an image are abnormal or not. This classifier is intended for use during the training and evaluation phases but not during inference. It defines a sequential neural network with three fully connected layers and ReLU activations, reducing the dimensionality from 1024 to 512 to 128 to 1 and sets up a binary cross-entropy loss function with a positive weight of 6.0 to counter the class imbalance, since abnormal regions are much less frequent than normal ones. This class takes “top_region_features”, “class_detected”, and “region_is_abnormal” – this is the ground truth of the regions’ abnormality – as inputs. Then, it computes the logits by passing “top_region_features” through the classifier and filters these logits to include only those corresponding to detected regions/classes. Lastly, it computes the binary cross-entropy loss for the detected regions using “region_is_abnormal” as ground truth. During training, it returns the computed loss while during evaluation, it returns both the loss and a boolean tensor indicating which regions are predicted to be abnormal. A threshold of -1 in logit space is used, meaning regions with logits > -1 are considered abnormal.

5. “binary_classifier_region_selection”

This class is designed to classify whether specific regions in an image should be selected for sentence generation. This class is used during both training and evaluation to improve the accuracy of sentence generation for selected regions. Additionally, a function “build_json” is included to update a JSON file with the classification results. Like the abnormality classifier, this classifier is composed of multilayer perceptron that compress the input features to a single output logit repeatedly and it sets up a binary cross-entropy loss function with a positive weight of 2.2 to address the class imbalance, since regions without sentences are more frequent than those with

sentences. It takes “top_region_features”, “class_detected”, “return_loss” – a boolean value indicating if the loss needs to be computed (necessary for training and evaluation) –, and “region_has_sentence” – the ground truth of the selection of the regions – as inputs. It also computes the logits by passing “top_region_features” through the classifier, computes the binary cross-entropy loss for detected regions using “region_has_sentence” as ground truth in the training and evaluation modes – it also uses a threshold of -1 in logit space –, filters out regions not detected by the object detector, and extracts “selected_region_features”, the features of the selected regions. During training, it returns the computed loss, while on evaluation mode, it returns the loss, “selected_regions”, and “selected_region_features” and during inference mode, it returns “selected_regions” and “selected_region_features”.

There is also a “build_json” function which reads the Json file that contain the coordinates of the region and add a “is_selected” key to each region and updates the same Json file but with the new key.

6. “llavamedmodel”

This class takes the Json file created in the previous step as an input. It then loads a quantized version of LLaVA-Med model. Firstly, the function “construct_prompt” constructs a prompt string with region analysis details to be used for generating the medical report. The function “generate_report” then iterates over each entry in the JSON data, extracts the image name and regions for each entry, constructs a prompt using “construct_prompt”, loads the corresponding image and converts it to RGB, processes the image and prompt into tensors, generates a report using the pretrained model by passing the processed inputs and specifying a maximum token length for the output, and finally, decodes the generated output into text, extracts the report part, and returns it.

7. “ReportGenerationModel”

This class integrates the object detector and the selection binary classifier and generate corresponding JSON files with bounding boxes and selection status for each region. It imports and loads pre-trained weights for the object detector and then uses it to get region features and detections, and then applies binary classifiers to these features to determine which regions are selected. Finally, it calls the “create_json_file” method to generate JSON output. This method constructs a JSON structure containing information about detected regions, including bounding boxes and whether each region is selected. The JSON structure groups regions into predefined sets (e.g., lung zones, hilar structures). This method returns the Json file as a string.

8. “generate_reports_for_images_llava”

In this file, the “ReportGenerationModel” and “llavamedmodel” are imported. There are a couple of functions in this file. “get_image_tensor” method reads and preprocesses a grayscale image, checks if the image is loaded correctly and ensures it's a single-channel image, applies a series of transformations, and converts the processed image to a tensor suitable for model input. “get_model” method loads a checkpoint containing the model's weights, loads these weights into a new instance of “ReportGenerationModel”, and sets the model to evaluation mode. So, the main workflow goes as follows: it loads the model, preprocesses the image, takes the Json file created earlier and generates the medical report using “llavamedmodel”. This file demonstrates an end-to-end pipeline from image input to report output.

4.3 User Interface

For our user interface, we wanted to implement a simple, user-friendly and easy to use interface to satisfy one of our objectives. We used a simple website to deploy our model. The website features a button that the user can click on to upload a chest x-ray image from the local machine. Afterward, the user clicks on another button to generate the report, which is displayed along with the uploaded image. When the user clicks on the “generate report” button, the image is passed into the machine learning pipeline used for inference, which generates text to be displayed to the user. The frontend of the website was made using HTML and CSS. And the backend of the website was made using Fast API, which is a straightforward way to host our website on our local machine. Figure 3 shows a screenshot of our user interface.

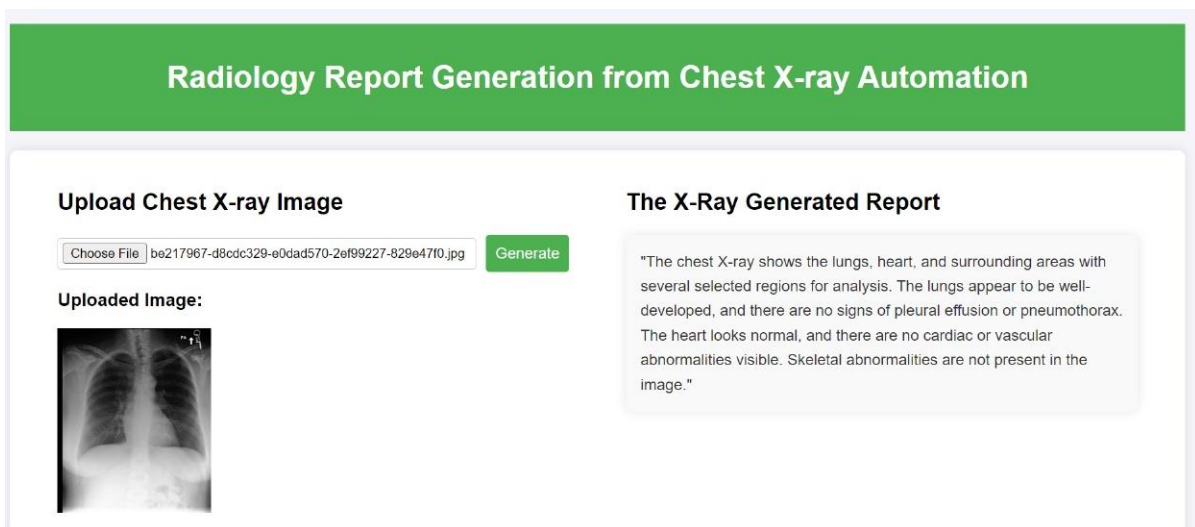


Figure 3: User Interface

4.4 Preliminary Experiments

Before integrating LLaVA-Med as our language model, we experimented with the LLaVA model by incorporating it into our workflow using LangChain and providing it with sample medical reports as references. LangChain was employed to manage the flow of data between the LLaVA model and our medical report generation system. It handled the input data, ensuring that the sample reports were properly formatted and fed into the model, and managed the outputs, attempting to tailor them to our specific needs. Despite these efforts, given that LLaVA is not pretrained on medical data and lacks familiarity with medical terminology, the generated reports fell short in terms of specificity and detail. The outputs remained broad and lacked the precision required for accurate medical documentation. This limitation highlighted the necessity of a specialized model like LLaVA-Med, which is tailored to understand and generate content rich in medical jargon and detailed descriptions. Our initial attempts with the standard LLaVA underscored the importance of domain-specific training, as the general model's outputs were insufficiently descriptive compared to the original medical reports, emphasizing the need for an advanced solution to meet the stringent demands of medical report generation.

Chapter 5

Results and Discussion

5.1 Quantitative Results:

Table 2 also shows the anatomy level results of the object detector presenting the intersection over union (IOU) values for each region.

Table 2: Anatomy level results

Anatomical Region	IOU
Right Lung	0.927
Right Upper Lung Zone	0.914
Right Mid Lung Zone	0.869
Right Lower Lung Zone	0.876
Right Hilar Structures	0.865
Right Apical Zone	0.849
Right Costophrenic Angle	0.664
Right Hemidiaphragm	0.771
Left Lung	0.916
Left Upper Lung Zone	0.919
Left Mid Lung Zone	0.869
Left Lower Lung Zone	0.847
Left Hilar Structures	0.855
Left Apical Zone	0.849
Left Costophrenic Angle	0.599
Left Hemidiaphragm	0.710
Trachea	0.783
Spine	0.858
Right Clavicle	0.803
Left Clavicle	0.767
Aortic Arch	0.646

Mediastinum	0.854
Upper Mediastinum	0.861
SVC	0.758
Cardiac Silhouette	0.816
Cavoatrial Junction	0.500
Right Atrium	0.702
Carina	0.149
Abdomen	0.904

Table 3 presents the results for the binary classifiers:

Table 3: Binary classifiers’ results

Metric	Result
Region Selection Precision	0.887
Region Selection Recall	0.650
Region Selection F1	0.886
Region Abnormality Precision	0.429
Region Abnormality Recall	0.905
Region Abnormality F1	0.513

Table 4 presents the Natural language generation (NLG) metrics for the full report generation task of our model in comparison with previous state-of-the-art models.

Table 4: NLG metrics for full report generation

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
PPKED[7]	0.360	0.224	0.149	0.106	0.149	0.284	0.237
MedXChat[11]	0.367	0.235	0.158	0.111	0.135	0.264	0.175
RGRG[12]	0.373	0.249	0.175	0.126	0.168	0.264	0.495
LLM-CXR[10]	0.196	0.095	0.054	0.033	0.081	0.245	0.445
LLaVA-Med[1]	0.232	0.086	0.027	0.009	0.082	0.168	0.015

Ours	0.241	0.176	0.121	0.093	0.181	0.195	0.471
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(Due to limited computational power our results are tested on a much smaller sample than the rest of the models)

To measure the clinical correctness of the generated reports we use the Clinical Efficacy (CE) scores using the conditions extracted from Chexbert[16], a BERT based information extraction system, to classify the status of 14 observations. We calculate the CE scores micro averaged over 5 observations, and calculate the CE scores example-based averaged over all 14 observations. Then we calculate the example-based precision, recall, and F1 scores over all 14 observations by comparing the classifications for each generated report and corresponding reference report. In addition to calculating the micro average precision, recall, and F1 scores over a subset of 5 observations: atelectasis, cardiomegaly, consolidation, edema, and pleural effusion. Table 5 shows the Clinical Efficacy (CE) metrics.

Table 5: CE metrics results

Metric	Result
Precision micro-5	0.254
Recall micro-5	0.319
F1 micro-5	0.283
Precision ex-14	0.239
Recall ex-14	0.246
F1 ex-14	0.231

5.2 Qualitative Results:

5.2.1 Object Detector

The object detector detects the 29 different regions inside the chest, splits them into five region sets, and outputs images that contain boxes marking each region in the chest as shown in figures 4 and 5.

The five region sets are:

```
region_set_1 = ["right lung", "right costophrenic angle", "left lung", "left costophrenic angle", "cardiac silhouette", "spine"]
```

```
region_set_2 = ["right upper lung zone", "right mid lung zone", "right lower lung zone", "left upper lung zone", "left mid lung zone", "left lower lung zone"]
```

```
region_set_3 = ["right hilar structures", "right apical zone", "left hilar structures", "left apical zone", "right hemidiaphragm", "left hemidiaphragm"]
```

```
region_set_4 = ["trachea", "right clavicle", "left clavicle", "aortic arch", "abdomen", "right atrium"]
```

```
region_set_5 = ["mediastinum", "svc", "cavoatrial junction", "carina", "upper mediastinum"]
```

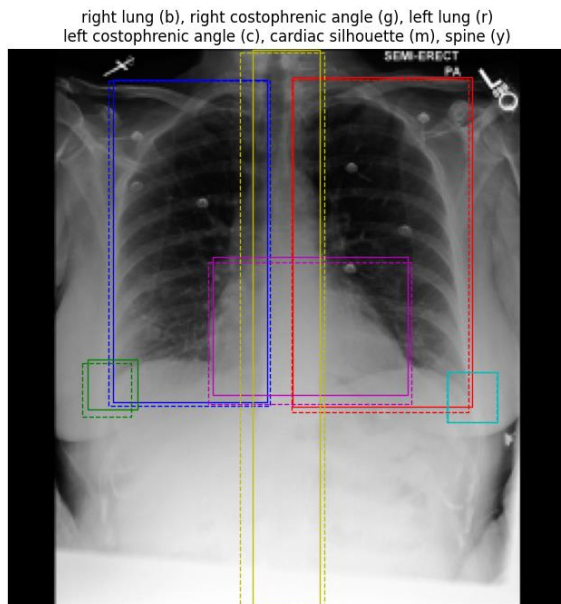


Figure 4: Object Detector Output 1

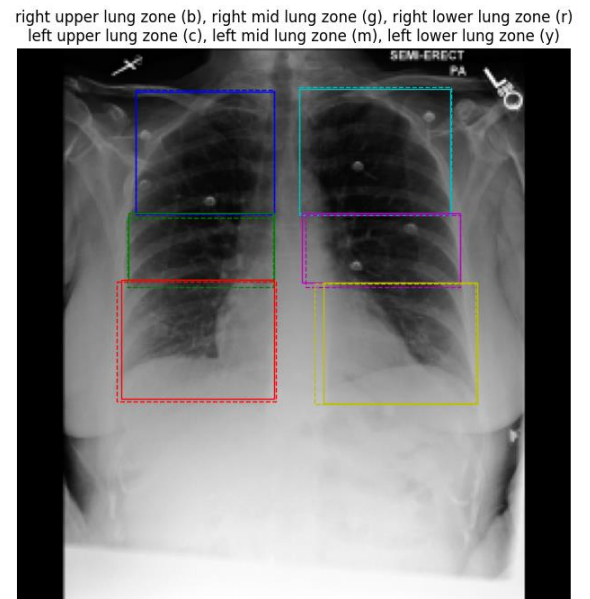
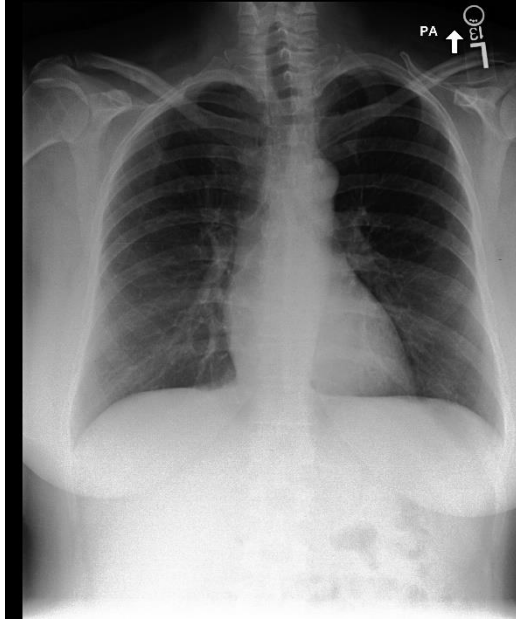


Figure 5: Object Detector Output 2

5.2.2 Report Generation:



Original Report:

“Frontal and lateral radiographs of the chest demonstrate normal heart size. The mediastinal and hilar contours are normal. No pleural effusion or pneumothorax. Clear lungs.”

Generated Report:

“The chest X-ray shows the lungs, heart, and surrounding areas with several selected regions for analysis. The lungs appear to be well-developed, and there are no signs of pleural effusion or pneumothorax. The heart looks normal, and there are no cardiac or vascular abnormalities visible. Skeletal abnormalities are not present in the image.”

Figure 6: Chest X-ray example

5.3 Discussion of Results:

Considering the IOU results of the object detector shown in table 1 and the difference between the marked boxes and the ground truth in figures 4 and 5, the object detection model is shown to accurately detected most of the anatomical regions inside the chest. However, few regions had lower values than the rest of the regions, which we expect is due to the sample size not being big enough, so the visual features of these specific regions were not clear in the images given to the model.

Regarding the report generation results; due to limited computational power we were not able to use the entire dataset in training and testing like previous state-of-the-art models

Despite being trained on a smaller sample of the dataset compared to previous state-of-the-art models due to limited computational power, our model performs comparably to them. However, it is anticipated that training the model on a larger portion of the dataset will likely yield even better results. Increasing the training data typically enhances the model's ability to generalize and capture more complex patterns in the data, leading to improved performance across various metrics. This approach is commonly employed to push the boundaries of model capabilities and achieve state-of-the-art results in machine learning research. Thus, future endeavors may focus on scaling up computational resources to capitalize on the potential gains from larger datasets.

Additionally, the METEOR score for our model showed the highest results compared to the other models mentioned, achieving a score of 0.181. This achievement can be attributed to the inherent strengths of the METEOR metric, particularly in the context of limited data. Unlike the other metrics, METEOR is known for its robust handling of recall and synonymy. The recall component ensures that the metric captures a higher proportion of relevant information from the reference texts, even when the training data is constrained. Furthermore, METEOR's ability to account for synonyms means that it can more effectively recognize semantically similar words, leading to a more accurate assessment of the model's output quality. These characteristics make METEOR especially suitable for scenarios where the dataset size is restricted, allowing our model to excel despite computational limitations.

Our model effectively captures the main findings for each region when compared to the original, utilizing a sophisticated integration of coordinates and inclusion criteria for each anatomical region. The language model is provided with coordinates generated by the object detector and informed by the selected region classifier on whether each region should be included in the report. This integration ensures that the data fed to the language model is comprehensive and precise, allowing it to generate accurate results. Even though there might be some variance in the terminologies used, the primary clinical findings remain relatively consistent and descriptive as shown in figure 6. The model's ability to interpret and synthesize this information highlights its robustness in handling complex medical data, demonstrating that it can maintain the integrity and detail of clinical findings. This method ensures that crucial information is not lost in translation, ultimately contributing to reliable and thorough medical reporting. The success of this approach underscores the potential for further enhancements, particularly by refining the model's terminology alignment and expanding the dataset, to ensure even greater accuracy and detail in future applications.

Chapter 6

Conclusion and Future Work

6.1 Limitations:

The limitations we faced during the project included:

1. Limited computational power and resources so we didn't have the ability to train the model on the entire dataset.
2. Our implementation solely focuses on chest x-ray only, due to its need for supervision.

6.2 Summary of Findings:

In summary, in this project we present a system that generates radiology reports from chest x-ray images using AI. This system features a region-guided approach that allows it to effectively capture relevant information in the chest x-ray image, by tending to each region in the chest area on its own. This allows a comprehensive approach that mimics the style of a radiologist assessing a chest x-ray. Afterward, a VLM is deployed and is given a specific prompt incorporating the region-guided information. This allows the VLM to generate comprehensive radiology reports that mimic radiologists' flow of writing. By incorporating advanced AI methodologies in the area of report generation, this project provides a system that generates relevant and consistent radiology reports and that can positively impact the healthcare industry.

6.3 Future Work:

Future improvements that could be made to our implementation may include training using a bigger sample of images, optimizing the code to make it more efficient and less computationally expensive, and enhancing the interpretability and explainability of the model's decisions by implementing techniques to provide visual and textual explanations for the detected regions and the generated report. Conducting clinical trials can also be done to validate the system's performance in real-world settings. In addition to gathering feedback from radiologists and other medical professionals to identify areas for improvement and ensure that the system meets clinical needs and standards. Areas for future research can include dataset expansion collecting more diverse datasets. Along with generalizing the implementation to different report generation tasks and extending the system to work with other medical imaging modalities, such as MRI, CT scans, and ultrasound images.

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